Jaya Sri Vardhan Samgoju

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GitHub | LinkedIn

PROFESSIONAL SUMMARY

Computer Science undergraduate with strong foundations in Artificial Intelligence, Machine Learning, Deep Learning, and Natural Language Processing, supported by hands-on experience through real-world academic and personal projects. Skilled in building intelligent systems using Python, FastAPI, PyTorch, Streamlit, and modern ML frameworks. Passionate about solving real-world problems through scalable software and AI-driven solutions, with a strong interest in Machine Learning, backend systems, and data-driven applications.

EDUCATION

B.Tech in Computer Science and Engineering — 2023 – Present
Rajiv Gandhi University of Knowledge Technologies (RGUKT), Ongole — **CGPA: 9.1 / 10**

Pre-University Course (PUC) — 2021 – 2023
Rajiv Gandhi University of Knowledge Technologies (RGUKT), RK Valley — **CGPA: 9.2 / 10**

Secondary Education (Class X)
Jawahar Navodaya Vidyalaya, Maddirala — **83%**

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C++
Machine Learning Deep Learning: Machine Learning, Deep Learning, CNN, RNN, LSTM, Transformers, Computer Vision, Anomaly Detection, Feature Engineering, Model Evaluation
NLP, LLM Agentic AI: Natural Language Processing (NLP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, Multi-Agent Orchestration, Prompt Engineering, Embeddings, Text Classification, Named Entity Recognition (NER), Vector Search
Frameworks Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, LangChain, LangGraph, Pandas, NumPy, OpenCV, MediaPipe
Backend MLOps: FastAPI, Docker, Apache Kafka, Streamlit, MLflow, EvidentlyAI, REST APIs, Git, GitHub
Databases Retrieval: FAISS, BM25, Vector Databases, Cross-Encoder Reranking

CERTIFICATIONS

- **Foundations of Deep Learning: Concepts and Applications** — NPTEL (Elite Certification, Score: 81%)
- **Complete Data Science, Machine Learning, Deep Learning & NLP Bootcamp** — Udemy
- **Generative AI and ChatGPT** — GeeksforGeeks
- **Ignite India Entrepreneurship Program** — Wadhwani Foundation

PROJECTS

SynoptiQ — Enterprise Agentic RAG & Document Intelligence System — *GitHub*

Stack: FastAPI ▪ LangChain ▪ LangGraph ▪ Groq (Llama-3.3-70B) ▪ FAISS ▪ BM25 ▪ CrossEncoder ▪ Next.js 14 ▪ TypeScript ▪ Docker ▪ JWT ▪ PyJWT ▪ Pydantic

- Built a **production-grade Agentic RAG system** using **LangGraph** to orchestrate a multi-agent pipeline (Planner → Retriever → Synthesizer → Verifier) capable of decomposing complex queries into targeted sub-queries and generating grounded, citation-backed answers from uploaded documents (PDF, DOCX, TXT).
- Engineered a **three-stage hybrid retrieval pipeline** combining **FAISS dense vector search**, **BM25 lexical retrieval**, and **cross-encoder reranking** (`ms-marco-MiniLM-L-6-v2`) to maximise both semantic and keyword recall, delivering top-K results with precision-optimised reranking.
- Implemented a **multi-layer guardrail system** with rule-based input and output validation to detect and block prompt injection, jailbreak attempts, and sensitive data leakage before queries enter or exit the agent pipeline, improving system security and compliance readiness.
- Developed a **Verifier Agent** using a zero-temperature Groq LLM call to evaluate groundedness and hallucination risk of every generated answer, returning structured confidence scores (0–1), verdict labels, and verification notes for full response traceability.
- Integrated **automated response quality evaluation** metrics per query including groundedness score, hallucination risk, retrieval quality, and answer completeness, alongside per-agent latency tracking and pipeline health reporting via an observability module.
- Built a **JWT-based authentication system with Role-Based Access Control (RBAC)** supporting Admin, Researcher, and Viewer roles, enforcing fine-grained permissions (`upload`, `query`, `manage_users`) via FastAPI dependency injection guards.
- Delivered a **real-time streaming endpoint** (`/query/stream`) using Server-Sent Events (SSE) for word-by-word response delivery,

alongside a full-stack **Next.js 14 frontend** with glassmorphism UI, agent trace panel, citation viewer, confidence badges, and markdown rendering.

- Authored **20+ unit test files** covering guardrails, JWT security, FAISS vector store, hybrid retrieval, agent orchestration, evaluation metrics, session memory, and document processing, ensuring end-to-end pipeline reliability.

VisionixAI — Zone-Based Computer Vision Automation *GitHub*

Stack: Python ▪ OpenCV ▪ MediaPipe ▪ Node.js ▪ Computer Vision ▪ CLI Automation

- Developed a **computer vision-based smart classroom automation system** capable of detecting student presence in predefined room zones and triggering automated ON/OFF actions without requiring additional hardware sensors.
- Designed a **zone-based occupancy detection pipeline** using **OpenCV** and **MediaPipe** to continuously monitor movement patterns, detect human presence, and identify occupancy duration within classroom regions.
- Implemented **real-time positional tracking logic** to analyze whether students remain within a specific zone for a predefined duration, enabling intelligent automation workflows based on sustained occupancy.
- Built a modular **Python-based ML core** for video stream processing, frame analysis, coordinate mapping, and zone-specific activity monitoring to support automated classroom management.
- Developed a **cross-platform CLI tool** using **Node.js** allowing users to analyze classroom video files through command-line execution, improving accessibility and ease of deployment.
- Engineered an automated decision system capable of **triggering zone-specific actions** based on visual presence detection, reducing dependency on traditional hardware sensors and manual monitoring.
- Optimized the system for **smart energy management and classroom automation**, supporting scalable deployment scenarios in educational and indoor monitoring environments.
- Structured the project with a modular architecture consisting of **CLI, ML Core, and automation workflows**, improving maintainability, debugging, and extensibility for future enhancements.

Real-Time Industrial Equipment Failure Detection System *GitHub*

Stack: Apache Kafka ▪ Isolation Forest ▪ FastAPI ▪ Streamlit ▪ MLflow ▪ EvidentlyAI

- Built a **real-time ML anomaly detection pipeline** ingesting high-frequency sensor telemetry streams via **Apache Kafka**, enabling scalable, low-latency processing of continuous industrial equipment data, enabling scalable, low-latency processing of continuous industrial equipment data.
- Developed a **sliding-window feature engineering framework** to extract statistical and temporal representations from streaming sensor signals, capturing operational trends, fluctuations, and early failure indicators for downstream anomaly detection.
- Implemented an **Isolation Forest anomaly detection model** for unsupervised identification of distributional outliers in unlabeled sensor data, enabling predictive equipment monitoring without dependency on labelled failure datasets.
- Engineered a **low-latency FastAPI inference service** exposing model predictions through modular REST endpoints, enabling seamless integration between the streaming telemetry layer, anomaly scoring, and downstream monitoring systems .
- Integrated **MLflow** for end-to-end experiment tracking, hyperparameter logging, and model versioning, supporting reproducible experimentation and structured comparison across model configurations.
- Deployed **EvidentlyAI** for continuous **data drift monitoring** on live sensor streams, enabling automated detection of distributional shifts and improving model robustness under evolving operating conditions.
- Built an interactive **Streamlit dashboard** for real-time visualisation of anomaly scores, drift reports, and sensor telemetry, supporting human-in-the-loop monitoring and operational interpretability for maintenance teams.

ACHIEVEMENTS

- Selected among the **Top 4 teams** in a Hackathon conducted at **KL University, Hyderabad** for developing a solution on the problem statement **"GST Reconciliation"**, focused on improving financial data validation and reconciliation processes.
- Successfully completed **NPTEL Elite Certification in Deep Learning** with a score of **81%**.
- Maintained strong academic performance with a **9.1 CGPA** in B.Tech Computer Science Engineering.

RELEVANT COURSEWORK

Machine Learning	Deep Learning	Natural Language Processing
Data Structures & Algorithms	Probability & Statistics	Database Management Systems